Factor-Anchored Reinforcement Learning for Portfolio Risk Control

Evidence from Frozen-Bundle Evaluation under Timing and Cost Constraints

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Preprint prepared in a NeurIPS-style conference template.

Abstract

This paper studies whether reinforcement learning is more useful in equity trading as a control layer than as an end-to-end predictor. The evaluated setup is a modular, factor-anchored pipeline in which finance-first alpha models feed a constrained allocator, while RL modulates exposure, protection, and execution under explicit timing and transaction-cost assumptions. Rather than relying on a best-case exploratory run, the paper ties its claims to a frozen archived bundle spanning April 2013 to March 2026. In the main archived split, the full pipeline reduces maximum drawdown relative to SPY by 9.8 percentage points (-24.0% versus -33.7%), but it trails SPY on annualized return (13.0% versus 15.3%) and Sharpe (0.53 versus 0.58). The factor-only benchmark and the strongest rule-based overlays remain materially stronger on return- and Sharpe-based comparisons, while differences versus end-to-end PPO are mixed and statistically inconclusive. The supported conclusion is therefore narrower still: in this archived bundle, RL is best viewed as an interpretable risk-shaping overlay rather than as a source of top-line performance or a clear win over end-to-end PPO.

Keywords: quantitative trading, reinforcement learning, factor investing, statistical arbitrage, regime detection, risk management, backtesting

1 Introduction

Many quantitative trading systems are built from a collection of specialized modules rather than from a single monolithic predictor. Cross-sectional factors, relative-value signals, volatility models, and regime filters often contribute distinct pieces of information about expected return and risk. Reinforcement learning (RL), by contrast, is most compelling when the task is inherently sequential: choosing when to increase risk, when to hedge, and how to adapt execution to changing conditions [1, 23].

This project adopts exactly that separation of responsibilities. The alpha layer is intentionally finance-first. It draws on ideas from factor investing [10, 15], cointegration-based statistical arbitrage [9], conditional volatility modeling [3, 8], and regime detection [13]. The RL layer does not attempt to discover alpha from scratch. Instead, it treats alpha as an input and learns how aggressively to express it under different risk and market states.

The present implementation contributes five practical elements to the evaluation setup.

1. A modular code architecture that cleanly separates data, alpha, control, backtesting, and visualization concerns.
2. A factor-anchored portfolio construction policy in which RL primarily controls exposure and active overlay size rather than acting as a second stock selector.
3. A constrained optimization layer between alpha and RL that turns signal forecasts into a capped, turnover-aware, and group-aware target book.
4. An IV-aware synthetic option-overlay sleeve and transaction-cost model that make the backtest more realistic while remaining implementable with public daily data.
5. A research evaluation workflow including ablations, rolling windows, sensitivity tests, and regime-conditioned policy summaries.

1.1 Research Questions

The current version of the report is organized around three explicit research questions.

1. **RQ1: Controller vs. end-to-end RL.** Is reinforcement learning more effective as a modular control layer over financially grounded alpha models than as an end-to-end trading agent?
2. **RQ2: Value over simple rules.** Does the RL control layer add measurable value beyond rule-based heuristics such as volatility targeting and drawdown-based deleveraging?

3. **RQ3: Component attribution.** Which part of the control layer, exposure modulation, dynamic hedging, or their joint operation, is the primary driver of the observed risk transformation?

Beyond the Present Manuscript

A natural next step, not evaluated in the present manuscript, concerns hedge realism: whether implied-volatility-conditioned, option-style overlays such as protective puts, collars, and put spreads improve the downside sleeve beyond the current stylized convex hedge approximation.

From a research perspective, this framing matters because it narrows the gap between a teaching prototype and a system that can be discussed in the language of quantitative portfolio construction. The objective is not to claim that the pipeline is production-ready. Rather, it is to assemble a well-motivated stack in which each modeling choice has a clear financial role and where performance can be decomposed into alpha generation, risk transformation, and control.

The central defended statement of the paper is therefore sharper than "the system works." *A factor-anchored RL control layer, evaluated under explicit cost and timing assumptions, can serve as a disciplined downside- shaping overlay relative to passive exposure while remaining more interpretable than an end-to-end policy. The paper does not claim that this RL layer surpasses SPY, PPO, or the strongest finance-first and rule-based baselines on return- and Sharpe-based comparisons in the archived bundle.* That narrower empirical outcome is still useful, because it helps separate where RL contributes to risk transformation from where simpler finance-first mechanisms remain stronger. The rest of the paper is organized to test that claim rather than merely to catalogue modules.

2 Related Literature and Design Motivation

The report is not meant to claim novelty in any individual modeling choice. Instead, it builds a coherent pipeline from well-established ingredients.

Factor investing. The cross-sectional component is aligned with the idea that common risk factors help explain returns [10]. Momentum is motivated by the medium-horizon continuation evidence of Jegadeesh and Titman [15], while quality and low-volatility proxies are included to stabilize the cross-section.

Statistical arbitrage. The pairs sleeve follows the classical cointegration logic of Engle and Granger [9]: if a spread is mean reverting around an equilibrium relation, deviations from that equilibrium may carry predictive content.

Volatility and regimes. Conditional heteroskedasticity motivates the GARCH sleeve [3, 8], while the market-regime component uses a two-state hidden Markov model in the spirit of Hamilton [13].

Reinforcement learning. RL is used as a controller rather than a raw alpha generator. This follows the practical observation that in finance, RL is often most useful for allocation, inventory, and execution decisions rather than for replacing domain-specific signal construction [1, 23]. This role assignment is also motivated by the low signal-to-noise ratio and limited effective sample size typical of financial data, which make end-to-end policy learning especially brittle.

Rule-based risk management and controller baselines. A key part of the evaluation design is to compare the RL controller not only with passive benchmarks but also with rule-based heuristics that a skeptical reader would reasonably propose as simpler substitutes. In this project, the most important examples are volatility targeting and drawdown-based deleveraging. These baselines matter because they help distinguish "RL adds value" from "any reasonable overlay would have helped."

2.1 Recent Work on RL for Portfolio Management

Recent work on RL in portfolio management and trading can be organized into four overlapping strands.

End-to-end policy learning. Early and influential work uses RL to learn trading or allocation policies directly from price histories and realized rewards [5, 6, 16, 19, 21, 24]. The appeal of this approach is conceptual elegance: the same learner is asked to extract signal, size positions, and adapt sequentially. The difficulty is that financial returns are noisy, nonstationary, and data-poor relative to many other machine-learning domains. As a result, end-to-end RL systems can be sample-inefficient, regime-sensitive, and hard to interpret when they fail [12, 14, 17].

Neural return forecasting with a separate optimizer. A second line of work uses deep or other machine-learning models to estimate returns, factors, or pricing kernels, then hands those forecasts to a conventional optimizer [4, 11, 12]. This preserves more structure in the allocation step than end-to-end RL, and often performs well empirically, but it still concentrates modeling risk in a data-hungry prediction layer.

RL for execution, hedging, and risk control. A third strand applies RL to narrower sequential tasks such as execution, hedging, and dynamic risk management rather

than to raw alpha discovery [2, 7, 18, 20]. This decomposition is closest to the present paper. The core idea is that RL may be most useful where the problem is inherently sequential and control-like, while alpha generation remains anchored to domain models or explicit financial structure.

Benchmarking and reproducibility. More recent platform and survey work has emphasized that RL in finance still suffers from weak evaluation discipline, narrow datasets, and inconsistent reporting [14, 22]. That critique directly motivates the frozen-bundle methodology used here. The paper is not only a claim about architecture; it is also a claim about how such architectures should be evaluated.

Table 1 positions the present approach relative to these broader families. The contribution is not a new RL algorithm. It is a methodological proposal: if strong finance-first alpha already exists, RL should be tested primarily as a controller layered on top of that alpha, then judged against both simple rule-based overlays and end-to-end learned alternatives.

2.2 Contribution Type

The contribution is not novelty in any single alpha ingredient. The paper does not claim a new factor model, a new GARCH variant, or a new hidden-state estimator. Its contribution lies instead in three places. First, it proposes a clear role assignment between alpha and control: financially grounded models produce expected-return structure, while RL handles sequential exposure and protection decisions. Second, it implements that role assignment in a modular architecture with an explicit constrained allocator between alpha and control. Third, it pairs the architecture with an evaluation workflow designed to test that claim via ablations, timing-discipline checks, robustness slices, and regime-conditional policy analysis. The contribution is therefore chiefly methodological: the paper proposes and tests a disciplined way to evaluate RL as a control layer rather than as a monolithic trading policy.

3 System Architecture

The pipeline is organized as a linear sequence of stages, illustrated in Figure 1. Each stage has a clearly delimited responsibility: upstream stages produce signals and risk features, the control layer translates these into portfolio decisions, and the backtest layer evaluates realized outcomes.

3.1 Software Structure

This decomposition is important for research reproducibility. It keeps the external command stable while turning the internals into separately testable components.

4 Data and Feature Construction

4.1 Universe

The archived research bundle uses the expanded universe rather than the smaller development basket. It contains 76 traded assets spread across technology, financials, healthcare, consumer staples, energy, industrials, materials, real estate, utilities, communications, and consumer discretionary, together with three diversifiers: GLD, TLT, and VNQ. SPY serves as the passive equity benchmark throughout.

4.2 Data Sources

The pipeline supports both a free baseline configuration and a richer research configuration. Daily prices and volumes are sourced via `yfinance`. When a `FRED_API_KEY` is available, macro data are drawn directly from FRED; in its absence, the system falls back to BLS and U.S. Treasury Fiscal Data. SEC company facts are incorporated when `SEC_USER_AGENT` is set, enabling a light fundamentals layer without requiring a proprietary data subscription.

4.3 Macro Regime Inputs

When available, the macro sleeve includes rates and labor-market variables such as the 10-year yield, 2-year yield, federal funds rate, unemployment rate, and term spread. The updated research design also incorporates stress-sensitive FRED series including the VIX, the ICE BofA high-yield option-adjusted spread, and a dollar-index proxy. These variables do not form a standalone alpha book. Instead, they help create a macro regime belief that is blended with the HMM regime estimate and provide additional context for risk-off states.

4.4 Uncertainty and Disagreement Features

One of the most distinctive methodological additions is that the controller is no longer conditioned only on point estimates. The updated state construction also tracks uncertainty proxies derived from the alpha and regime stack. The main examples are alpha-sleeve dispersion, regime-belief entropy, and rolling instability in recent information coefficients. Intuitively, these features measure three dif-

Table 1: Taxonomy of portfolio-management architectures.

Approach	Alpha source	Control logic	Interp.	Representative references
Classical factors	Hand-crafted financial signals	Fixed allocator or simple overlay rules	High	Fama–French style factor investing [10]
End-to-end RL	Learned from market histories	RL policy directly selects positions or weights	Low to medium	Direct trading and portfolio RL [5, 16, 19, 24]
Neural forecasting + optimizer	Neural or ML return forecasts	Separate optimizer translates forecasts into holdings	Medium	Asset-pricing and return-forecast models [4, 11, 12]
This paper	**Finance-first alpha models**	**RL modulates exposure, hedging, and execution**	**High**	**Frozen-bundle control-layer evaluation**

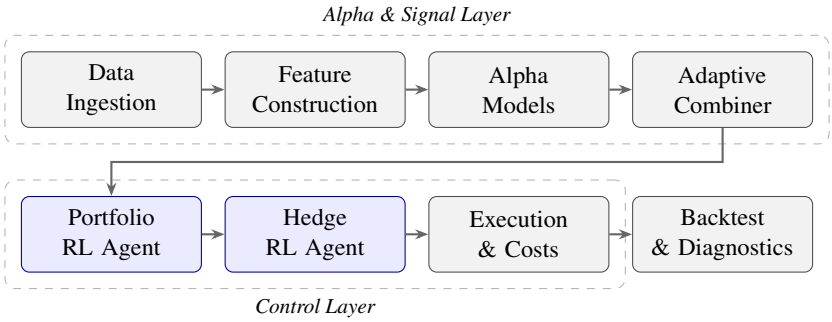

Figure 1: High-level architecture of the quantitative trading pipeline. The alpha and signal layer (top) produces expected-return scores and risk features; the control layer (bottom) translates these into portfolio weights, hedge intensity, and cash allocation before evaluation in the backtest engine.

ferent kinds of uncertainty: disagreement across signals, ambiguity in the inferred market state, and instability in the recent mapping from scores to realized returns.

This matters because a sequential controller should not react the same way when the alpha opportunity looks strong and coherent as when the same opportunity is accompanied by high model disagreement or unstable recent signal quality. In that sense, the control layer is not just regime-aware; it is also uncertainty-aware.

4.5 Information-Timing Contract

One of the most important research upgrades in the current version is that the pipeline now makes its feature-availability assumptions explicit. Macro inputs are shifted forward by a configurable number of trading days before they are allowed into the backtest. This is still weaker than a true real-time vintage database, because it does not model revisions, but it is materially stronger than using the final macro history with no delay assumption at all.

Fundamental inputs are handled more conservatively. The SEC quality signal is still a static cross-sectional research prior rather than a fully point-in-time fundamentals panel. For that reason the research engine now supports a strict timing mode in which the SEC quality sleeve is disabled and macro inputs are forced to use the largest configured lag. The point of this design is not to pretend that point-in-time fundamentals are already solved, but rather to make the timing caveat measurable and visible in the evaluation workflow.

5 Methodology

5.1 Problem Formulation

Let $r_{t+1} \in \mathbb{R}^N$ denote next-period asset returns for a universe of N tradable assets. At time t, the system observes a feature state $x_t \in \mathcal{X}$ constructed from cross-sectional factor values, relative-value spreads, volatility forecasts, regime beliefs, and macro inputs. The alpha layer maps these features into an expected-return score vector $\alpha_t \in \mathbb{R}^N$ and a confidence vector $c_t \in \mathbb{R}^N$.

The control layer then chooses a portfolio allocation w_t, a cash allocation, and a hedge intensity h_t. The one-step portfolio objective is not written as a pure return maximization problem. Instead, it is implicitly multi-objective: maximize long-run wealth while limiting drawdown, volatility, and unnecessary turnover. This is the reason the project separates alpha estimation from control: the two tasks operate on different time scales and carry different inductive biases. Importantly, the RL layer does

Table 2: Module-level decomposition of the `quant_stack` package.

Module	Responsibility
`config.py`	Universe definition, global constants, and external series identifiers
`data.py`	Market price/volume ingestion, macro series (FRED / BLS fallback), and SEC company facts
`alpha.py`	Cross-sectional factors, pairs cointegration, GARCH volatility, HMM regime, LSTM forecast, and adaptive signal combination
`rl.py`	Portfolio construction agent, execution controller, and dynamic hedging agent
`pipeline.py`	Walk-forward backtest orchestration and online RL update loop
`evaluation.py`	Ablation, rolling-window robustness, sensitivity analysis, and research artifact export
`plots.py`	Equity-curve, drawdown, RL behavior, and alpha-model diagnostic figures
`main.py`	CLI entry point and compatibility wrapper

not forecast r_{t+1} directly; it maps a state summarizing alpha quality and risk conditions into exposure, overlay, and protection decisions.

5.2 Cross-Sectional Factor Model

The factor sleeve combines four z-scored components: momentum, a value proxy, a quality proxy, and a low-volatility tilt. Letting m_i, v_i, q_i, and ℓ_i denote the standardized scores for asset i, the composite factor score is

$$f_i = 0.30m_i + 0.25v_i + 0.25q_i + 0.20\ell_i.$$

This choice reflects a balanced, multi-style factor book rather than a single directional bet.

5.3 Pairs Trading

For candidate pairs, a hedge ratio β is estimated by OLS and the spread

$$s_t = P_t^{(1)} - \beta P_t^{(2)}$$

is normalized into a z-score

$$z_t = \frac{s_t - \mu_s}{\sigma_s + \varepsilon}.$$

Large positive z-scores correspond to short-spread trades and large negative z-scores correspond to long-spread trades.

5.4 Volatility and Regime Models

Each name receives a GARCH(1,1) forecast

$$\sigma_t^2 = \omega + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2,$$

where $\omega > 0$, $\alpha_1 \geq 0$, and $\beta_1 \geq 0$ are the intercept, ARCH, and GARCH coefficients respectively. This conditional variance estimate is used for confidence scaling and risk awareness. Separately, a two-state HMM produces a bull-versus-bear regime belief. The final regime signal is

$$\text{RegimeBelief}_t = 0.75 \cdot \text{HMMBelief}_t + 0.25 \cdot \text{MacroBelief}_t.$$

5.5 Adaptive Alpha Combination

Let $s_{i,t}^{(k)}$ denote the signal from source k for asset i at time t. The expected return score is formed as

$$\alpha_{i,t} = \sum_k w_t^{(k)} s_{i,t}^{(k)},$$

where the source weights $w_t^{(k)}$ adapt over time based on recent realized signal quality. In the implementation, this is proxied by recent Spearman information coefficients between source scores and realized returns.

This adaptive weighting is important because the constituent alpha sleeves are not equally informative across all market states. Pairs signals can be weak in strongly trending markets, deep nonlinear forecasts can degrade when regime structure shifts, and factor signals can vary in efficacy depending on the dispersion of the cross-section. The adaptive combiner therefore behaves like a simple online model-selection layer.

5.6 Portfolio Construction RL

The portfolio RL state summarizes the current cross-sectional alpha opportunity, recent realized portfolio volatility, the prevailing regime belief, and an uncertainty score built from alpha dispersion, regime entropy, and recent IC instability. Crucially, the policy is *factor-anchored*. The target book is

$$w_t^{\text{target}} = 0.90\, w_t^{\text{factor}} + 0.10\, w_t^{\text{stabilizer}},$$

where the stabilizer is a long-only minimum-variance-style portfolio estimated from a Ledoit–Wolf shrinkage covariance matrix.

If the RL agent chooses invested fraction b_t and active overlay size τ_t, the final invested portfolio is approximately

$$w_t = b_t \left[(1-\tau_t)w_t^{\text{target}} + \tau_t w_t^{\text{satellite}}\right].$$

The satellite sleeve is not a fully independent book; it tilts the factor target toward the strongest alpha names. This keeps the full pipeline close to the working factor engine while still letting RL modulate aggressiveness.

Between the alpha combiner and the RL policy, the code now inserts a constrained intermediate allocator. The optimizer is long-only and solves a small regularized problem of the form

$$\min_{w} \quad \lambda_{\text{risk}} w^\top \Sigma_t w - \lambda_\alpha \alpha_t^\top w + \lambda_{\text{anchor}} \|w - w_t^{\text{target}}\|_2^2 + \lambda_{\text{turnover}} \|w - w_{t-1}\|_2^2$$

subject to $w_i \geq 0$, $\sum_i w_i = 1$, per-asset caps, and simple group concentration limits. This layer is architecturally important because it makes the separation of concerns sharper: alpha expresses expected return, the optimizer expresses institutional portfolio constraints, and RL governs when to lean into or away from that target book. This intermediate layer is not only an architectural convenience; it reflects the practical fact that many real portfolio systems mediate between forecast signals and final positions through a constrained allocator.

5.7 Reward Design

The portfolio RL update uses a differential Sharpe-style reward based on the realized portfolio return relative to a recent rolling distribution of returns. In compact form,

$$R_t^{\text{portfolio}} = \frac{r_t - \hat{\mu}_t}{\hat{\sigma}_t + \varepsilon},$$

where $\hat{\mu}_t$ and $\hat{\sigma}_t$ are running estimates of recent mean return and volatility. This means the agent is not rewarded solely for taking more risk in absolute terms; it is rewarded for improving return relative to the recent volatility environment.

The hedge agent uses an asymmetric realized-return reward in which losses are penalized more heavily than gains are rewarded. A stylized form is

$$R_t^{\text{hedge}} = r_t \mathbf{1}(r_t \geq 0) + \lambda_{\text{loss}} r_t \mathbf{1}(r_t < 0), \qquad \lambda_{\text{loss}} > 1.$$

That asymmetry is a deliberate design choice: it pushes the hedge layer toward preserving capital in stress rather than maximizing average carry.

The research runner is now also designed to support reward-function ablations. In addition to the default differential-Sharpe-style objective, the evaluation protocol can compare raw-return and Sortino-style reward formulations. This matters because reward design is one of the least observable but most important degrees of freedom in RL-for-finance systems. Because reward specification changes the controller's operating point, it should be treated as a methodological degree of freedom rather than as a neutral implementation detail.

5.8 No-Trade Band and Hedging

To reduce turnover drag, a rebalance band suppresses small changes in target weights. When only part of the portfolio is changed, the active weights are rescaled to preserve the desired capital budget.

The hedging RL layer observes drawdown, volatility regime, recent momentum, and IV-aware option-overlay features. In the archived experiment reported here, it chooses both one of four hedge intensities and one proxy overlay type among a protective put, a collar, and a put spread. The total one-period return can be written schematically as

$$r_t^{exttotal} = r_t^{extrisky}(1 - \phi h_t) + r_t^{extcash} + r_t^{exthedge},$$

where h_t is the hedge fraction and $r_t^{exthedge}$ includes carry, option-style payoff asymmetry, and a state-dependent protection term.

This is more realistic than the earlier convex-loss proxy, but it is still not a market-priced options backtest. The overlay payoffs are generated from public IV and market-stress proxies rather than live option-chain quotes. The reported hedge sleeve should therefore be interpreted as a synthetic option-style overlay rather than as a fully market-consistent derivatives implementation.

5.9 Transaction Costs and Execution Realism

The backtest now includes a three-part execution-cost approximation. For each rebalance, the strategy pays a fixed basis-point fee on turnover, an additional penalty proportional to turnover times realized volatility, and a size penalty linked to the largest one-name weight change. In symbols,

$$\text{TC}_t = c_0 \cdot \text{turnover}_t + c_1 \cdot \text{turnover}_t \cdot \hat{\sigma}_t + c_2 \cdot \max_i |\Delta w_{i,t}|.$$

This is not a full execution simulator, but it pushes the daily allocator away from unrealistic churn and makes the rebalance band economically meaningful rather than purely cosmetic.

6 Experimental Design

The archived bundle used in the paper contains approximately 3,270 daily observations from early April 2013 through late March 2026. The reported main split trains the upstream statistical components on the initial segment through September 25, 2019 and then evaluates the walk-forward controller from September 25, 2019 to March 27, 2026. The manuscript is tied to this frozen archived bundle, saved on March 28, 2026 as the latest complete research run for the expanded-universe configuration, even though the live codebase may already contain extensions not exercised by the reported evidence. This does not eliminate prior exploratory iteration, but it does make the reported bundle explicit and reproducible. Within the archived suite itself, action ladders, optimizer settings, reward choices, and comparison baselines are held fixed across ablations, split-fraction tests, and rolling-window robustness slices.

Within that archived run, the LSTM sleeve is first trained on the initial window, after which the full pipeline is rolled forward one trading day at a time with the configuration held fixed during evaluation.

At each date, the pipeline executes the following sequence. First, factor, pairs, GARCH, HMM, macro, and LSTM features are generated from the available data. These are combined into expected-return scores and confidence levels via the adaptive combiner. The portfolio RL agent then selects an exposure bucket and overlay size, after which the hedge agent chooses both a hedge intensity and a proxy option-overlay type. Transaction costs, cash carry, and hedge payoffs are applied to arrive at the realized portfolio return, which is finally used to update both RL value tables before advancing to the next trading day.

The comparison set now goes beyond passive and naive portfolio baselines. In addition to SPY buy-and-hold and the factor-only benchmark, the research protocol includes a volatility-targeting overlay on the factor book, a drawdown-based deleveraging rule, and an end-to-end PPO baseline that receives the same broad information set as the modular architecture. These additions are essential for the paper's research questions because they test whether RL adds value over simple heuristics and whether modular control is preferable to a monolithic learning policy.

6.1 Benchmark Logic

The benchmark set is chosen to isolate different sources of value. SPY measures whether the pipeline adds value relative to standard passive equity exposure. The factor-only benchmark isolates the pure alpha layer without any RL overlay. Because the archived bundle already uses a broad 76-asset universe, this factor benchmark is also informative about what the diversified opportunity set can do without RL control. Comparing the full pipeline against the factor-only benchmark is particularly important because it reveals whether RL is adding value as a controller or merely acting as a drag on a strong signal engine.

The newer comparison baselines sharpen that logic further. The volatility targeting baseline asks whether most of the observed improvement can be matched by simple inverse-vol scaling. The drawdown-based deleveraging baseline asks whether a threshold rule can replicate the protection sleeve's effect. The end-to-end PPO baseline addresses the paper's most distinctive architectural question: if one hands the same state information to a monolithic policy, does it outperform a hybrid design in which finance-specific modules produce the target book and RL only controls how aggressively to express it? These baselines are deliberately strong because they represent the kinds of simple overlays a practitioner would try before accepting the complexity of RL.

7 Evaluation Metrics

The reporting suite includes annualized return, annualized volatility, Sharpe ratio, Sortino ratio, maximum drawdown, Calmar ratio, and return-distribution diagnostics including a daily 5% Value-at-Risk estimate.

If $\{r_t\}_{t=1}^{T}$ denotes daily portfolio returns and r_f is the annual risk-free rate, then the annualized Sharpe ratio is

$$\text{Sharpe} = \frac{252\,\bar{r} - r_f}{\sqrt{252}\,\hat{\sigma}(r) + \varepsilon}.$$

The Sortino ratio replaces total volatility with downside volatility, while the Calmar ratio scales annualized return by the magnitude of maximum drawdown. Each metric emphasizes a different aspect of the return distribution, which matters in this project because the pipeline is explicitly designed to reshape tail risk, not merely to maximize mean return. Table 3 focuses on the primary comparison metrics used in the main strategy table: return, volatility, Sharpe, maximum drawdown, and Calmar. Sortino and Value-at-Risk are retained as secondary diagnostics in the text, reward-ablation evidence, and archived research artifacts.

In addition, the research runner emits regime-conditional summaries of portfolio action, hedge ratio, cash weight, turnover, transaction-cost drag, and realized returns. These diagnostics help evaluate whether the con-

troller behaves differently in bullish, neutral, and bearish states in a way that is consistent with the design intent of the policy.

For statistical reliability, the updated evaluation design also includes block-bootstrap confidence intervals on daily-return paths. The motivation is to preserve some of the short-horizon dependence structure in returns while still providing uncertainty bands for quantities such as annualized return, Sharpe, Calmar, and pairwise performance differences versus benchmark strategies.

8 Results

This section moves from the main walk-forward evidence to ablation, comparisons against rule-based overlays and end-to-end PPO, rolling robustness slices, blocked time-series cross-validation, and regime-conditioned behavior. The organizing question throughout is whether the data support the narrower claim that RL helps more as a controller than as a predictor.

8.1 Main Result

The archived research run used for the paper's main evidence evaluates the walk-forward controller from September 25, 2019 to March 27, 2026. The summary metrics are shown in Table 3. This is still only one realized path, but it is the same frozen checkpoint that also feeds the ablation, robustness, and significance artifacts discussed in the remainder of the paper.

The main result is now more clearly mixed than favorable. That shift matters, because it forces the paper to defend only the claims that survive the expanded universe and longer sample.

Relative to SPY, the full pipeline is weaker on annualized return, 13.0% versus 15.3%, and slightly weaker on Sharpe, 0.53 versus 0.58. Its favorable result is narrower: maximum drawdown improves by about 9.8 percentage points, from -33.7% for SPY to -24.0% for the full pipeline, and Calmar improves modestly from 0.45 to 0.54. The passive-benchmark comparison therefore supports downside reshaping, but not broad outperformance.

Relative to the factor-only benchmark, the conclusion is decisively negative. The full pipeline trails by roughly 13.6 percentage points of annualized return and by about 0.47 of Sharpe. The factor core remains the strongest pure alpha expression in the archived bundle.

Relative to the strongest rule-based overlays, the result is again negative. Volatility targeting and drawdown-based deleveraging both beat the full pipeline on return and Sharpe, and they do so with drawdowns that are shallower than, or at least comparable to, the RL stack.

Relative to the end-to-end PPO baseline, the result is mixed rather than favorable. The full pipeline earns a higher annualized return, 13.0% versus 8.7%, but slightly lower Sharpe, 0.53 versus 0.56, and materially worse maximum drawdown, -24.0% versus -15.5%. The main table therefore does not provide a clean modular-over-PPO win on point estimates.

Implication for RQ1 and RQ2. The main table is mostly unfavorable for the RL layer. Drawdown improvement relative to SPY survives, but neither superiority to strong rule-based overlays nor a clear advantage over end-to-end PPO is supported.

8.2 Ablation Result

The most important workshop-level question is not whether the full system does well on one path, but whether the component layering supports the paper's architectural claim. For that reason the research runner now emits an ablation summary comparing the factor-only benchmark, the alpha stack without RL, alpha plus portfolio RL, alpha plus hedge RL, and several reduced-state full-pipeline variants across multiple split settings.

The generated ablation table sharpens the story considerably, and more negatively than before. Averaged across split settings, the factor-only benchmark remains the strongest pure alpha expression, with mean return 18.7% and mean Sharpe 0.82. The pairs and LSTM sleeves add almost no incremental value on top of that base and slightly worsen average drawdown. The alpha stack without RL stays close to the factor core on Sharpe, 0.81, while improving mean maximum drawdown to -24.8%.

Among the RL variants, the best result comes from portfolio RL alone, not from the hedge sleeve or the full stack. Alpha plus portfolio RL reaches mean Sharpe 0.79, the best mean drawdown in the table at -21.7%, and the highest mean Calmar among the RL configurations. By contrast, alpha plus hedge RL falls to mean Sharpe 0.49, and the full pipeline to 0.43.

The additional state features are also not earning their keep in this archived bundle. Removing regime, uncertainty, or volatility state inputs slightly improves the full pipeline in every case. This is a stronger pruning result than the earlier bundle: in the current archived evaluation, neither the hedge sleeve nor the richer RL state description justifies its added complexity.

Implication for RQ3. Not supported in the intended direction. The current ablation evidence does not show hedge RL as the main value driver; if anything, the least-

Table 3: Performance summary on the walk-forward test window (25 Sep 2019 – 27 Mar 2026). Annualized figures. Max DD denotes maximum peak-to-trough drawdown; Calmar is annualized return divided by |Max DD|.

Strategy	Return	Volatility	Sharpe	Max DD	Calmar
Full pipeline	13.0%	17.8%	0.53	−24.0%	0.54
SPY	15.3%	20.2%	0.58	−33.7%	0.45
Factor-only benchmark	26.6%	23.0%	1.00	−33.8%	0.79
Vol-target	16.3%	14.3%	0.89	−20.6%	0.79
DD-Delever	19.3%	17.1%	0.92	−21.4%	0.90
E2E RL (PPO)	8.7%	9.2%	0.56	−15.5%	0.56

Table 4: Ablation summary across train/test split settings.

Component	Return	Vol.	Sharpe	Max DD	Calmar
Factor-only benchmark	18.7%	18.3%	0.82	−26.9%	0.72
Factor + pairs	18.3%	18.3%	0.80	−27.1%	0.69
Factor + LSTM	18.5%	18.3%	0.81	−27.1%	0.70
Alpha stack (no RL)	17.9%	17.5%	0.81	−24.8%	0.73
Alpha + portfolio RL	16.4%	16.1%	0.79	−21.7%	0.75
Alpha + hedge RL	12.4%	17.5%	0.49	−26.0%	0.47
Full pipeline	10.7%	16.3%	0.43	−22.3%	0.47
Full (no regime)	11.5%	16.2%	0.47	−22.7%	0.50
Full (no uncert.)	10.9%	16.2%	0.44	−23.1%	0.46
Full (no vol)	11.3%	16.4%	0.46	−24.5%	0.45

damaging RL contribution comes from portfolio exposure control alone.

8.3 Rule-Based and End-to-End Baseline Evidence

The next layer of evidence asks whether the RL controller is doing something meaningfully different from simple heuristics. On this archived run, the volatility-targeting and drawdown-based deleveraging baselines are deliberately strong comparators: both deliver meaningfully higher return- and Sharpe-based performance than the full pipeline. The bootstrap and Jobson–Korkie tests point in the same direction. Relative to the factor-only benchmark, the full pipeline's Sharpe deficit is clearly negative. Relative to vol-target and drawdown-based deleveraging, the deficits are borderline to unfavorable, remaining negative in point estimate and near conventional significance thresholds.

In the archived sample, RQ2 is not supported. The evidence in this manuscript does not support the claim that RL adds value over the strongest simple rule-based overlays on Sharpe or return.

RQ1 receives only weak support at best. The end-to-end PPO baseline earns lower nominal return than the full pipeline, but slightly higher Sharpe and materially shallower drawdown. The bootstrap Sharpe delta versus PPO is -0.03 with a wide confidence interval, and the Jobson–Korkie comparison is likewise far from significant. The appropriate conclusion is therefore not that modular RL beats PPO, but that neither RL formulation establishes a convincing edge over finance-first or rule-based alternatives, while the modular architecture remains easier to interpret and decompose.

Implication for RQ1. Not strongly supported. The modular design improves annualized return relative to PPO, but not Sharpe or drawdown, and the risk-adjusted differences are not statistically decisive.

Implication for RQ2. Not supported in the archived sample: the strongest simple non-RL overlays outperform the RL controller on return and Sharpe.

8.4 Reward-Function Sensitivity

Reward design affects where the controller lands on the risk-return frontier, but now only within a fairly narrow band. Across the archived reward-function ablation, the Sortino-style reward is best, reaching 13.8% annualized return and 0.59 Sharpe, while the default differential-Sharpe reward reaches 12.9% and 0.53. Mean-variance is the weakest on drawdown-adjusted terms, with maximum drawdown worsening to −25.5%. Reward choice therefore matters, but it does not rescue the controller from the broader ranking.

The archived reward-ablation evidence is therefore spe-

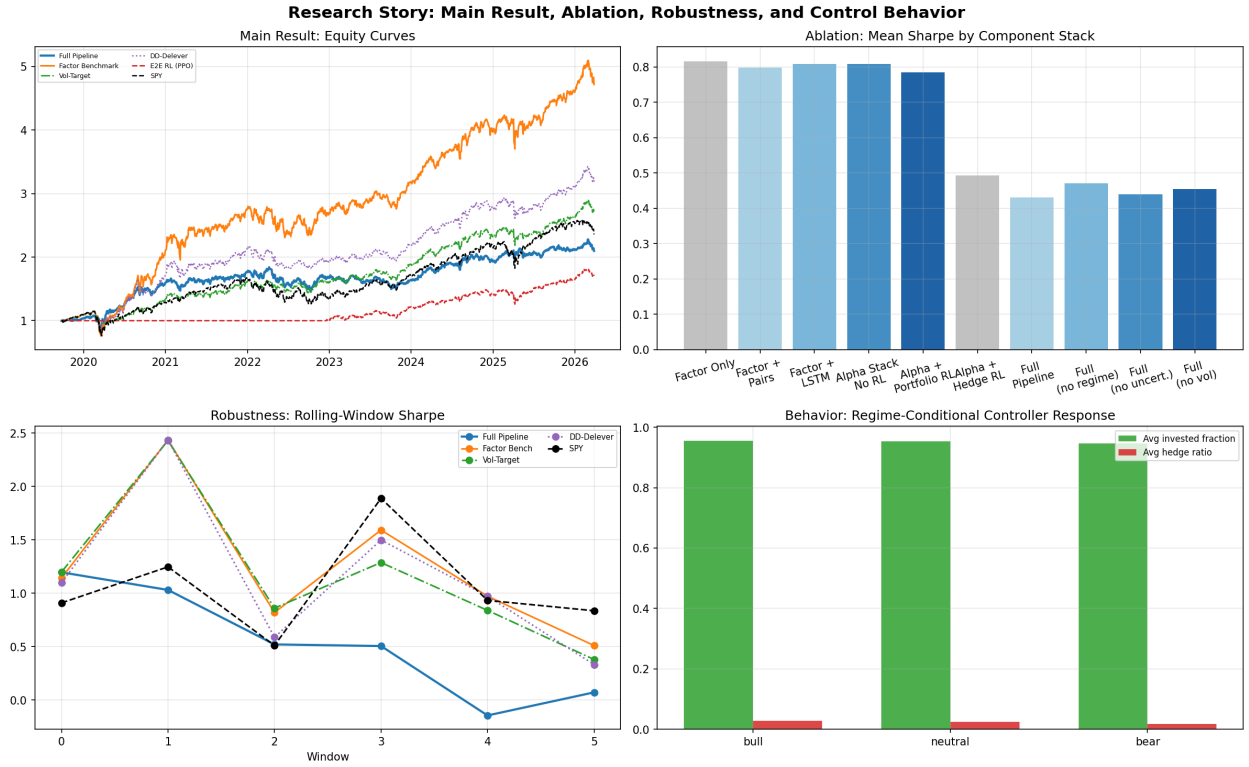

Figure 2: Compact workshop-style summary figure generated by the robustness engine. The four panels show the main equity-curve result, ablation evidence, rolling-window robustness, and regime-conditioned controller behavior. The figure makes clear that the strongest raw-performance outcomes in the archived bundle come from the factor-only benchmark and the strongest simple overlays, and that the RL-based variants remain well below those comparators on top-line metrics.

cific to the modular full pipeline. A comparable PPO reward sweep is not treated as primary evidence in this manuscript because the archived suite did not preserve an equally complete reward-by-split comparison grid for PPO, and the smaller set of PPO reward runs is materially noisier because each trial depends on fresh stochastic policy training.

Implication for methodology. Reward choice matters substantially for where the controller lands on the risk-return frontier, but it does not overturn the paper's main qualitative result: reward tuning changes the controller's operating point within a weak overall band without overturning the broader qualitative ranking in which the factor-only benchmark and the strongest simple overlays remain stronger on return- and Sharpe-based comparisons.

8.5 Robustness Across Windows

Single-path backtests can be persuasive but are rarely sufficient. The research runner therefore evaluates the full pipeline across rolling windows and blocked time-series cross-validation. In the archived run, the rolling-window picture is materially weaker than in the earlier March 27 bundle: median full-pipeline Sharpe is 0.51, median Calmar 0.83, and median maximum drawdown -10.9%. The full pipeline beats SPY on Sharpe in only 33% of windows and on Calmar in only 17% of windows. Relative to the factor-only benchmark, it matches or improves drawdown in half the windows but never exceeds factor-only on Calmar.

Blocked time-series cross-validation reinforces that instability. Across five folds, annualized Sharpe ranges from 0.11 to 1.22, with one strong fold and several much weaker ones. That dispersion suggests substantial regime dependence rather than a stable control advantage. These windows and folds still come from one historical sample and should therefore be interpreted as internal robustness slices rather than independent replications.

The same evidence also sharpens the paper's statistical tone. The bootstrap and Jobson–Korkie summaries show that the full pipeline is not statistically distinguishable from SPY or PPO on Sharpe in the archived sample, while its Sharpe deficit relative to the factor-only benchmark is clearly negative and its deficits relative to volatility targeting and drawdown-based deleveraging are borderline to unfavorable. The appropriate claim, then, is not that ro-

Table 5: Rolling-window robustness summary for the full pipeline.

Statistic	Value
Rolling windows	6
Median full-pipeline Sharpe	0.51
Median full-pipeline Calmar	0.83
Median full-pipeline Max DD	−10.9%
Frac. windows full beats factor-only benchmark on drawdown	50%
Frac. windows full beats factor-only benchmark on Calmar	0%
Frac. windows full beats SPY on Sharpe	33%
Frac. windows full beats SPY on Calmar	17%

Table 6: Block-bootstrap significance for full pipeline versus baselines (Sharpe deltas).

Comparison	Delta Sharpe	95% CI	p-value
Full – SPY	-0.05	[-0.71, 0.45]	0.830
Full – Factor-only benchmark	-0.47	[-0.86, -0.14]	0.000
Full – Vol-target	-0.36	[-0.78, 0.04]	0.075
Full – DD-Delever	-0.39	[-0.83, 0.02]	0.070
Full – E2E RL (PPO)	-0.03	[-1.12, 0.91]	0.950

Table 7: Reward-function ablation for the full pipeline on the archived bundle.

Reward	Return	Sharpe	Max DD	Calmar
Differential Sharpe	12.9%	0.53	−24.2%	0.53
Return	13.1%	0.54	−23.7%	0.55
Sortino	13.8%	0.59	−23.7%	0.58
Mean-Variance	13.1%	0.55	−25.5%	0.51

bustness is "won," but that the architecture now supports a falsifiable and transparent protocol for showing when RL does *not* generalize.

Implication for the overall claim. The rolling and cross-validation evidence do not support a structural edge. At most, they support occasional drawdown improvement relative to passive exposure.

8.6 Behavioral and Regime Evidence

The behavioral question is whether the controller is acting like a sequential decision system. Figure 2 provides the main summary used in the paper. The answer in the current archived bundle is only weakly positive. The controller is not degenerate: it does hold cash, change turnover, and sometimes select protective-put overlays. But the regime-conditioned summaries do not show the intended monotone mapping from worse states to lower exposure and stronger protection.

In the representative full-pipeline split, average portfolio action rises from about 1.92 in bull states to 2.15 in bear states, while average hedge ratio falls from roughly 4.9% to 2.7%. The dominant hedge type is a protective put in bull states and no hedge in neutral and bear states. That pattern is hard to interpret as rational downside control. The more plausible reading is that the current regime labels, state features, and hedge policy are not yet aligned tightly enough to produce reliable state-contingent behavior.

Implication for RQ3. Not supported in the intended behavioral sense. The controller is active, but its hedge and exposure choices are not yet consistently regime-rational.

9 Discussion

The experimental results support three conclusions of methodological importance, but they are more negative than the previous archived bundle.

First, the data and alpha side of the pipeline is doing most of the credible work. The factor-only benchmark remains the strongest pure alpha engine, and the alpha stack without RL stays close on Sharpe while improving drawdown somewhat. The pairs and LSTM sleeves add little incremental value. In the current archived evaluation, the strongest evidence is for the factor core and the constrained non-RL allocator, not for the full RL stack.

Second, the RL layer is currently defensible only in a narrow downside-shaping sense. In the main split it reduces drawdown relative to SPY, but it does not beat

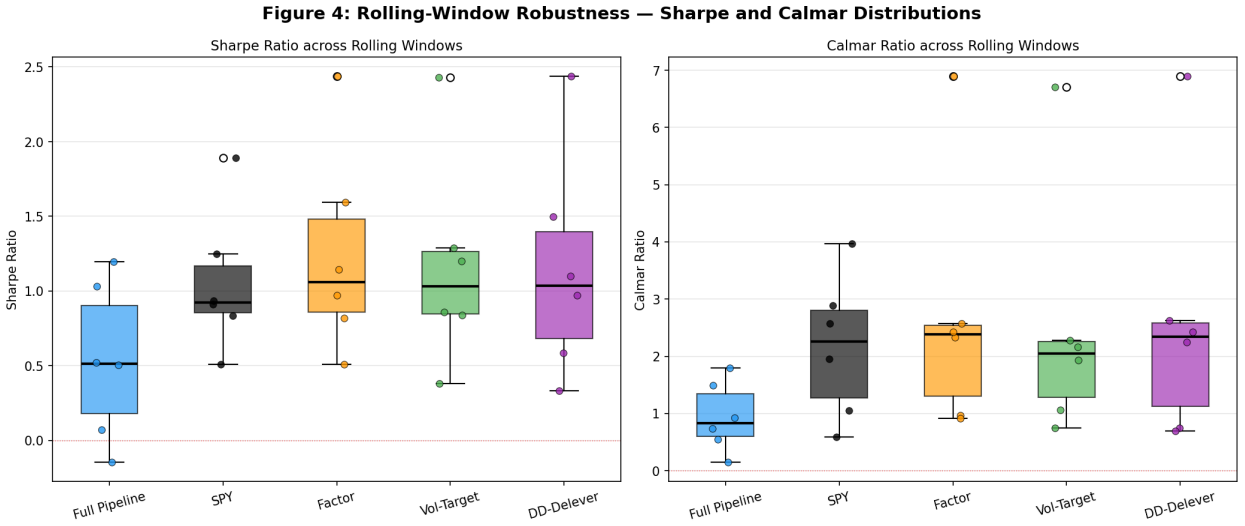

Figure 3: Rolling-window robustness from the archived March 28, 2026 experiment bundle. The full pipeline shows mixed performance across windows, beating SPY on Sharpe in only one-third of slices and never exceeding the factor-only benchmark on Calmar. These rolling windows are internal robustness slices from one archived historical sample rather than independent replications.

SPY on return or Sharpe, does not beat the factor-only benchmark or the best rule-based overlays, and does not produce a clean modular-over-PPO win. That is still a useful empirical outcome: it shows that the right question is not "does RL win?" but "what, exactly, does RL change once a finance-first signal engine is already in place?"

This distinction is central. If the RL layer were treated as a second stock selector, it would risk duplicating the alpha problem with a less structured model and less data efficiency. By anchoring the policy to the factor book, the pipeline preserves interpretability: alpha determines *what* to own, while RL primarily determines *how much* to own and *what overlay* to carry. The problem is not that the control formulation is incoherent; it is that the current controller is not yet adding enough economic value over simpler alternatives.

What did not work is equally important. RL did not beat the strongest rule-based overlays on Sharpe or return. The hedge sleeve underperformed the simpler portfolio-RL-only variant. The richer regime, volatility, and uncertainty state inputs did not improve the full pipeline relative to stripped- down versions. The regime-conditioned behavior is not monotone in the intended direction. These are useful negative results because they narrow where future improvements should be sought. For a control-layer study, these failures are not incidental; they are precisely what helps delimit where RL is and is not adding value.

The practical lesson is therefore more about simplification than expansion. Further gains are more likely to come from pruning weak sleeves, redesigning the hedge logic, and validating state features against stronger non-RL baselines than from adding more controller complexity. True option quotes, cleaner regime definitions, and less noisy controller objectives may matter more than richer state spaces by themselves.

9.1 When Practitioners Should Use This Architecture

The archived evidence suggests a narrower use case than the paper's original prototype narrative. This architecture is most defensible as a research or governance-friendly framework when interpretability and explicit downside shaping matter more than maximizing Sharpe, and when a credible finance-first alpha engine already exists underneath the controller.

Simpler alternatives are often sufficient in other settings. If the primary goal is pure risk-adjusted performance on a daily U.S. equity allocation problem, the archived bundle suggests that the factor-only benchmark and simple overlays such as volatility targeting or drawdown-based deleveraging are harder to beat. If computational budgets are tight, rule-based overlays are also easier to train, debug, and explain. And if the alpha layer is weak, RL is unlikely to rescue the system: the controller can reshape risk, but it cannot compensate for the absence of robust signal quality.

The practical contribution is therefore not "always use RL." It is narrower still: when RL is used in finance, the current bundle suggests treating it as an interpretable research instrument for exposure and protection decisions, not yet as a superior production allocator.

10 Threats to Validity and Future Work

The implementation remains research code, and several caveats apply. Macro inputs are shifted by a configurable lag, but they are not full vintage series with revision-aware histories. SEC enrichment can be used as a static cross-sectional prior, but it still does not constitute a true point-in-time fundamentals database. Transaction costs are more realistic than before, yet they remain stylized rather than venue-specific. The hedge sleeve in the archived experiment uses synthetic option-style overlays driven by IV proxies rather than live option-chain quotes, and the robustness engine still depends on one underlying historical sample even when it slices that sample in multiple ways.

Several additional threats deserve explicit acknowledgement. First, the design has been iterated by observing backtest behavior, which raises the possibility of researcher degrees of freedom even when each individual change is motivated. Second, the traded universe is broader than in the earlier prototype but still narrow relative to institutional equity universes, so the results may depend partly on the selected names. Third, daily data are appropriate for a medium- horizon allocation study but cannot validate intraday execution claims. Fourth, the hedge sleeve is a proxy options implementation rather than a market-priced options backtest. Finally, the project deliberately leans on free or public data sources, which improves accessibility but necessarily limits point-in-time fidelity and market microstructure realism.

A further caveat is benchmark fairness: although the rule-based and PPO baselines are designed to be strong and informative, their exact specification still influences comparative conclusions. Because the evaluation suite reports many metrics and comparator strategies, some apparent wins and losses should also be interpreted with the usual caution around multiple comparisons. External validity is also limited: conclusions drawn from a daily U.S.-equity allocation setting may not transfer unchanged to other asset classes, frequencies, or market microstructure regimes.

The currently reported robustness summary is still based on only six rolling windows, five blocked time-series CV folds, and one underlying historical sample. The workshop interpretation should therefore be that the robustness engine is now part of the methodology, not that robustness has been exhaustively established across many independent market environments.

Natural next steps include true filing-date-aware fundamentals, release-vintage macro datasets, cross-asset regime instruments, and a market-priced hedge sleeve using live option surfaces rather than IV proxy payoffs. A second important line of follow-up work is empirical pruning: the LSTM sleeve, the pairs sleeve, and the richer RL state variables should remain in the architecture only if ablation continues to show incremental value after further robustness passes. Finally, while the evaluation engine now includes bootstrap significance and Jobson–Korkie outputs, future work could push further toward additional universes, stronger multiple-testing discipline, and deeper validation across disjoint market regimes.

11 Conclusion

This paper shows that a financially grounded hybrid architecture can be modular, interpretable, and empirically informative even when its RL layer is not a performance winner. Classical quantitative models remain responsible for alpha discovery, while reinforcement learning acts as a sequential risk and allocation controller. The resulting system is not a proof that RL dominates traditional quant methods. Instead, it is a structured research pipeline whose behavior is interpretable, testable, and falsifiable.

The main empirical takeaway is narrower and more negative than in the earlier bundle: in the March 28, 2026 archived evaluation, the modular controller reduces drawdown relative to SPY in the main split, but it does not beat SPY on return or Sharpe, does not clearly beat end-to-end PPO, and remains well behind the factor-only and best rule-based baselines. Equally importantly, the paper leaves behind a frozen-bundle evaluation protocol that can support future pruning, hedge redesign, and market-priced options extensions under a consistent empirical standard. In the reported sample, the evidence supports RL chiefly as an interpretable risk-shaping overlay and rejects stronger claims of current economic superiority over finance-first or simple rule-based alternatives.

A Implementation Snapshot

A.1 Archived Experiment Control Ladders

The March 28, 2026 archived experiment bundle used in the main paper results relies on the discrete action ladders below. Unlike the earlier bundle, the reported archived evaluation already includes implied-volatility-conditioned synthetic hedge-type selection among protective puts, collars, and put spreads. What remains future work is replacing these proxy payoffs with market-priced option data.

Portfolio action	Invested fraction	Active overlay fraction
State 0	82%	5%
State 1	90%	8%
State 2	95%	12%
State 3	98%	16%
State 4	100%	20%

Hedge action	Hedge ratio
State 0	0%
State 1	3%
State 2	8%
State 3	15%

Hedge type action	Overlay type
Type 0	Protective put
Type 1	Collar
Type 2	Put spread

A.2 Research Artifacts

For the archived run discussed in the paper, the research runner writes the artifacts below into `../../results/20260328_experiments/` alongside the standard paper outputs:

1. `research_metrics.csv`: tabular output for ablation and robustness runs;
2. `research_ablation_summary.csv`: split-aggregated ablation summaries;
3. `research_robustness_summary.csv`: rolling-window medians and benchmark beat fractions;
4. `research_regime_summary.csv`: regime-conditioned action, hedge, turnover, and return summaries;
5. `research_rolling_references.csv`: rolling benchmark metrics for SPY and the factor-only benchmark;
6. `research_bootstrap_cis.csv`: block-bootstrap confidence intervals for key strategy metrics;
7. `research_bootstrap_significance.csv`: pairwise bootstrap deltas and p-value summaries versus benchmark strategies;
8. `research_jobson_korkie.csv`: parametric Sharpe-ratio comparison tests;
9. `research_ts_cv.csv`: blocked time-series cross-validation summaries;
10. `research_summary.json`: serialized configuration and best-per-suite summaries; and
11. `research_paper_tables.tex` together with `pipeline_research_eval.png`: paper-ready summary tables and figure.

Together with the feature-availability contract embedded in each backtest run, these artifacts make it easier to compare results across alternative timing, cost, and control assumptions without manually reconstructing the experimental context after the fact.

A.3 Supplementary Diagnostic Figures

These figures are diagnostic and illustrative; the primary evidentiary claims of the paper rest on Tables 3–6 and Figures 2 and 3. The appendix intentionally focuses on figures produced directly by the frozen March 28, 2026 bundle and omits older alpha or execution dashboards that are not tied to the reported evidence.

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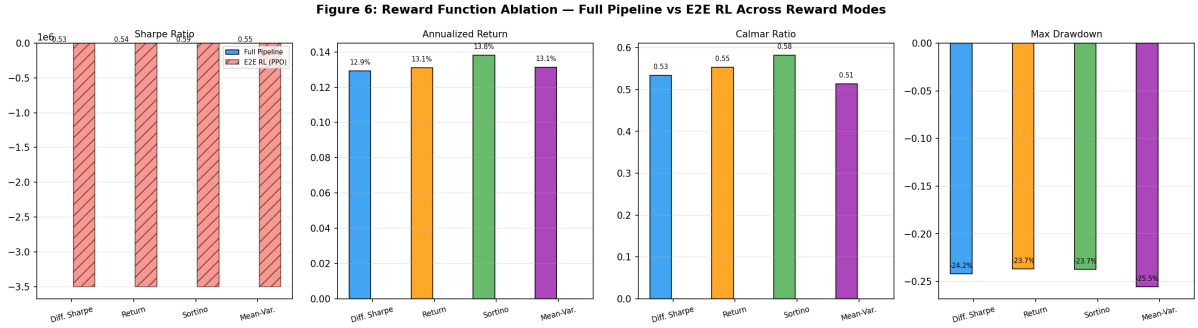

Figure 4: Supplementary reward-ablation figure for the archived bundle. Reward choice moves the full pipeline within a relatively narrow performance band, and no reward variant changes the broader ranking in which the factor-only benchmark and the strongest rule-based overlays remain stronger on return- and Sharpe- based metrics.

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